EE1101: Circuits and Network Analysis - Quiz 4

Name: ______________________________

Roll No: ______________________________

Instructions

1. The exam duration is 45 minutes, starting at 15:40. Ensure that you return the paper by 16:15. Please be aware that **late submissions** may incur **penalties**.
2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.
3. Adopting unfair practices will lead to an F grade and will be reported to the disciplinary committee.

I hereby declare that I have read all the instructions carefully and will adhere to them.

. **Signature:** ____________________

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1. (2 points) Compute the maximum value of inductor current for the circuit shown in Fig. 1. Assume that t_0 is large enough for the circuit to operate in steady state before t_0. Assume $i_l(0) = 0$ and $v_c(0) = 0$. Enter your answer here

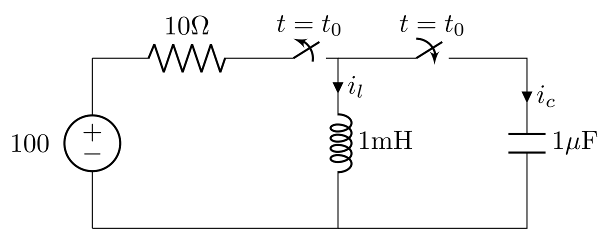

Fig. 1: Circuit for question 1

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2. (8 points) Consider the circuit shown in Fig. 2.

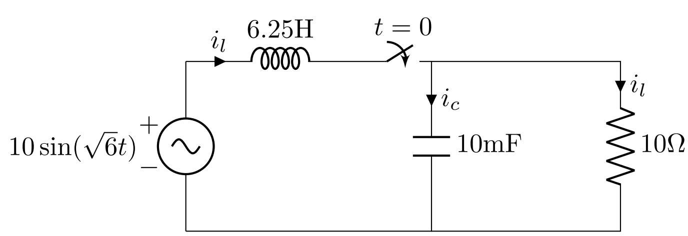

Fig. 2: Circuit for question 2

(a) (2 points) Compute the complete response (including the transient part) of the voltage across the resistor. Enter the expression in the space below. (Make sure that it is clearly legible.)

(b) (2 points) Compute the steady state voltage across the resistor in phasor form. Enter your answer here: ________________.

(c) (2 points) It is desired to steady state voltage express the voltage across the capacitor in the form $v_c(t) = V_m \sin(\sqrt{6}t + \phi)$. Compute the value of V_m and ϕ. Enter the value of V_m here ________________ and ϕ here: ________________.

(d) (2 points) Let $\mathbf{S}_s$ and $\mathbf{S}_r$ denote the complex power associated with source and resistor respectively. Compute $\mathbf{S}_s + \mathbf{S}_r$. Enter the value here: ________________.

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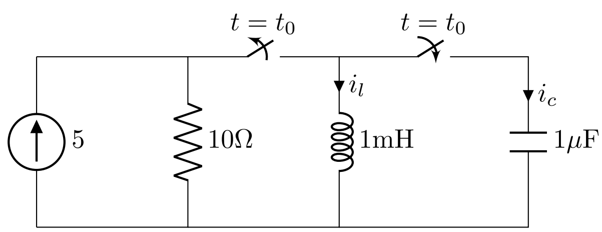

Fig. 1: Circuit for question 1

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2. (8 points) Consider the circuit shown in Fig. 2.

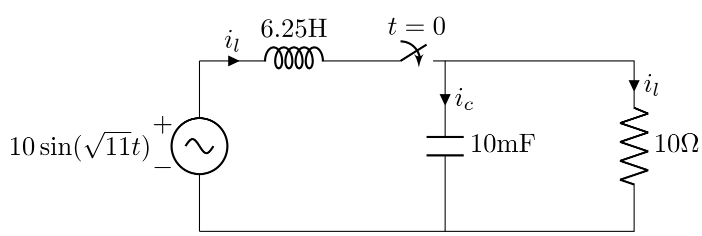

Fig. 2: Circuit for question 2

(a) (2 points) Compute the complete response (including the transient part) of the voltage across the resistor. Enter the expression in the space below. (Make sure that it is clearly legible.)

(b) (2 points) Compute the steady state voltage across the resistor in phasor form. Enter your answer here: ________________.

(c) (2 points) It is desired to steady state voltage express the voltage across the capacitor in the form $v_c(t) = V_m \sin(\sqrt{11}t + \phi)$. Compute the value of V_m and ϕ. Enter the value of V_m here ________________ and ϕ here: ________________.

(d) (2 points) Let $\mathbf{S}_s$ and $\mathbf{S}_r$ denote the complex power associated with source and resistor respectively. Compute $\mathbf{S}_s + \mathbf{S}_r$. Enter the value here: ________________.

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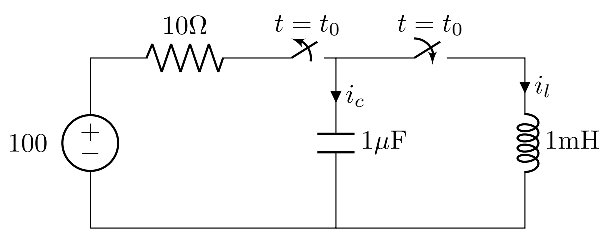
1. (2 points) Compute the maximum value of capacitor voltage for the circuit shown in Fig. 1. Assume that t_0 is large enough for the circuit to operate in steady state before t_0. Assume $i_l(0) = 0$ and $v_c(0) = 0$. Enter your answer here



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2. (8 points) Consider the circuit shown in Fig. 2.

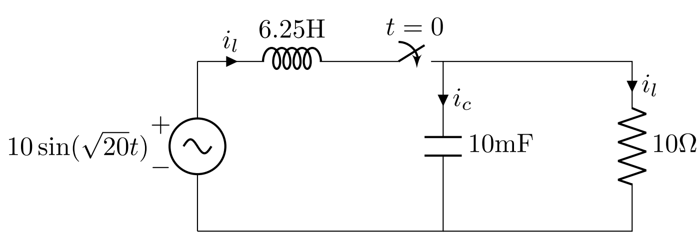

Fig. 2: Circuit for question 2

   (a) (2 points) Compute the complete response (including the transient part) of the voltage across the resistor. Enter the expression in the space below. (Make sure that it is clearly legible.)

(b) (2 points) Compute the steady state voltage across the resistor in phasor form. Enter your answer here: ________________.

(c) (2 points) It is desired to steady state voltage express the voltage across the capacitor in the form $v_c(t) = V_m \sin(\sqrt{20}t + \phi)$. Compute the value of V_m and ϕ. Enter the value of V_m here ________________ and ϕ here: ________________.

(d) (2 points) Let $\mathbf{S}_s$ and $\mathbf{S}_r$ denote the complex power associated with source and resistor respectively. Compute $\mathbf{S}_s + \mathbf{S}_r$. Enter the value here: ________________.

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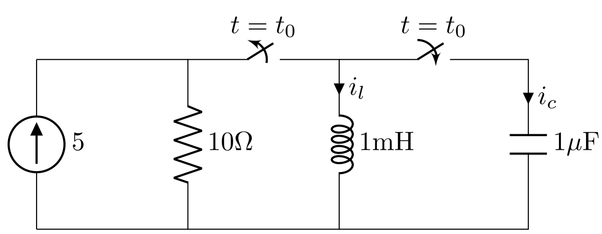
1. (2 points) Compute the maximum value of capacitor voltage for the circuit shown in Fig. 1. Assume that t_0 is large enough for the circuit to operate in steady state before t_0. Assume $i_l(0) = 0$ and $v_c(0) = 0$. Enter your answer here



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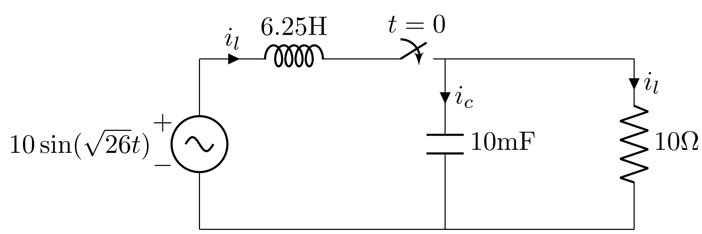

Fig. 2: Circuit for question 2

(a) (2 points) Compute the complete response (including the transient part) of the voltage across the resistor. Enter the expression in the space below. (Make sure that it is clearly legible.)

(b) (2 points) Compute the steady state voltage across the resistor in phasor form. Enter your answer here: ________________.

(c) (2 points) It is desired to steady state voltage express the voltage across the capacitor in the form $v_c(t) = V_m \sin(\sqrt{26}t + \phi)$. Compute the value of V_m and ϕ. Enter the value of V_m here ________________ and ϕ here: ________________.

(d) (2 points) Let $\mathbf{S}_s$ and $\mathbf{S}_r$ denote the complex power associated with source and resistor respectively. Compute $\mathbf{S}_s + \mathbf{S}_r$. Enter the value here: ________________.